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The longitudinal effects of global and regional brain measurements on cognitive abilities

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Loss of brain tissues and cognitive abilities are natural processes of aging, but longitudinal studies are limited. We explored the longitudinal association between global and regional brain measures with cognitive abilities among individuals with normal cognitive status, mild cognitive impairment, and Alzheimer's Disease, including those with stable or progressive cognitive status, from the Alzheimer's Disease Neuroimaging Initiative. Follow-up was conducted approximately every six months for two to three years to classify participants by their cognitive status (*n* = 814). Mixed effects models were used to measure the longitudinal association between global and regional brain measures with cognitive abilities assessed by the Alzheimer's Disease Assessment cognition sub-scale after controlling for covariates. While whole brain volume (controlling for head size) affected the rate of cognition changes among those with mild cognitive impairment or Alzheimer's, other volumetric and thickness measures (hippocampus volume, ventricles volume, and entorhinal cortex thickness) had differential effects on cognition, but not on the speed of their changes. Results highlight the potential importance of different brain regions on cognition, but perhaps not on the rate of change and should be further investigated.

Keywords Alzheimer's disease, Cognitive abilities, Global and regional structural brain, Cognitive impairment

Background

Aging naturally results in biological changes (such as decreased brain volume)[1]. Alzheimer's Disease (AD) is a progressive brain disease that can accelerate memory, thinking, and behavioral problems[2]. Although people with AD live for 3 to 10 years after diagnosis[3], research indicates that the brain starts changing years before the onset of neurodegenerative biomarkers[4]. Thus, due to this latency period[5], it is critically important to identify and treat people when they are demonstrating symptoms of mild cognitive impairment (MCI)[6], rather than when they have progressed to AD. Studying this phase improves our understanding of early stages of AD and supports intervention development to slow its progression. However, not everyone with MCI will progress to AD. Indeed, some will revert to normal cognitive health, and others will remain MCI[7]. This heterogeneity[6] underscores the importance of identifying characteristics which differentiate between those who remain stable with MCI, and those who progress to AD.

Demographic and health differences (such as education level and having the apolipoprotein E-ε4 allele) between those who remain MCI, revert to normal, and convert to AD have been reported[8–10] but notable limitations of the existing longitudinal literature exist. For instance, most longitudinal studies have a small number of follow-up visits (e.g., two time points over one or two years)[11,12] which not only decreases the power and precision of the results[13] but also limits the ability to detect changes in the predictors over a longer period of time[14]. Moreover, some of the previous studies used the Mini-Mental State Examination (MMSE) to assess the global change in cognitive skills[12]. However, the MMSE is a screening test and is inadequate in identifying changes over time[2].

Lastly, unmeasured confounding may be a limitation of many previous studies. For instance, volumetric measures of the brain including the whole brain volume (WBV), hippocampus volume (HCV), and ventricles

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volume (VEV) as well as the entorhinal cortex thickness (ERC) are reportedly significantly different between MCI and cognitively healthy people[15–17]. Given this heterogeneity, investigating potential interactions between brain volumetric measures and cognitive abilities over time is needed. Thus, the objectives of this study were to (1) examine the longitudinal effect of WBV on cognitive abilities and changes in those abilities over time across groups with different cognitive status, (2) examine the longitudinal effect of regional brain volume (HCV, VEV) and ERC on cognitive abilities and changes in those abilities over time across groups with different cognitive status.

Methods

Data for this study were from the Alzheimer's Disease Neuroimaging Initiative (ADNI) longitudinal cohort (https://adni.loni.usc.edu/). The ADNI was launched in 2003 as a public-private partnership, led by Principal Investigator Michael W. Weiner, MD. The primary goal of ADNI has been to test whether serial magnetic resonance imaging (MRI), positron emission tomography (PET), other biological markers, and clinical and neuropsychological assessment can be combined to measure the progression of mild cognitive impairment (MCI) and early AD. For up-to-date information, see www.adni-info.org.

Participants from ADNI1 (2004–2010) with brain MRI data ($n = 819$) were used for this study. The inclusion criteria were: English or Spanish speaking adults between 55 and 90 years of age, and with no comorbidities which might interfere with study participation[6]. Participants were followed up approximately every 6 months for 2–3 years. Written informed consent was obtained from all participants; ADNI was approved by the institutional review boards of all participating institutions. Research was conducted in accordance with the Declaration of Helsinki. As multiple universities and institutions are heavily involved with ADNI data collection and study management, their full details are omitted from this manuscript. They can be found in the ADNI protocol and ADNI consent forms on the publicly accessible ADNI website: https://adni.loni.usc.edu/methods/documents/. This secondary data analysis was approved by Concordia University's Human Research Ethics Committee (#30019564). De-identified data are publicly available upon approval from the ADNI Data Sharing and Publications Committee.

Measures

Cognitive status

At baseline and each follow-up visit, the cognitive status of participants was assessed with validated questionnaires, brain imaging, and a physician's assessment. Validated questionnaires included: Logical Memory II sub-scale score from the Wechsler Memory Scale (test-retest = 0.75), Mini-Mental State Examination (MMSE) score (Cronbach's $\alpha = 0.78$), and Clinical Dementia Rating (Cronbach's $\alpha = 0.83$)[18–20]. In accordance to the literature's guidelines[18], the ADNI protocol classified participants into three mutually exclusive cognitive groups based on these measures (Logical Memory II, MMSE, and Clinical Dementia Rating): cognitively normal (CN): no symptoms of cognitive impairment or dementia; MCI: no subjective memory concerns or signs of dementia but reported difficulties with daily activities, and AD: based on meeting clinical guidelines[18]. CN participants were age-matched with MCI and AD participants. Of the total ADNI1 participants at baseline, 229 were classified as CN, and 398 and 192 participants were classified as MCI and AD, respectively.

In this study, participants were classified as 'stable-cognitively normal' if their cognition classification at follow-up was the same as that at baseline. Participants were classified as 'progressors' if their cognition classification worsened over time. If participants progressed to a status that was different than their baseline status and then reverted to the baseline status, they were classified as 'unstable'. Thus, six groups were identified: (1) stable-cognitively normal (sCN, n = 209), (2) stable-MCI (sMCI, n = 206), (3) those who progressed from cognitively normal to MCI, (pCN, n = 18), (4) those who progressed from MCI to AD (pMCI, n = 171), (5) MCI participants who progressed and reverted (unsMCI, n = 20), and (6) those who had AD at baseline and remained AD through the rest of follow-up (AD, n = 190). Due to small numbers, CN participants who progressed and reverted (e.g., cognitively normal at baseline, converted to MCI at a follow-up visit, and reverted to cognitively normal at a subsequent follow-up visit) were excluded from the study (n = 4). One subject was also eliminated from the study due to missing data at all time-points, therefore our final analytic sample was 814 participants.

Assessment of cognitive abilities over time using ADAS-Cog13

The modified Alzheimer's Disease Assessment Scale-cognition sub-scale (ADAS-Cog13; Cronbach's $\alpha = 0.55$)[21,22] was used in this study to measure cognitive abilities (i.e., person's performance on cognitively demanding tasks)[23] and to study the changes in these abilities over time[23]. This measure was collected at every visit, with scores ranging from 0 to 85, with higher scores indicating greater cognitive dysfunction. In ADNI1, the ADAS-Cog13 was not used during screening or follow-up visits to determine participants' cognitive group (e.g., CN, MCI).

Brain measurements

As indicated in the literature, whole brain volume, hippocampal volume, ventricular volume, and entorhinal cortex thickness serve as key biomarkers in cognitive neuroscience studies. For instance, some studies showed that these biomarkers differ significantly between individuals with mild cognitive impairment (MCI) and cognitively healthy controls[15,16]. Research also suggests that cortical thickness is a promising biomarker for detecting brain atrophy, as it is more sensitive to layer-specific changes than volumetric methods and benefits from the ability to be automatically estimated, which enhances its applicability in large-scale studies[24]. Therefore, Entorhinal cortex thickness (ERC), and volumetric measures such as whole brain volume (WBV), hippocampus volume (HCV), and ventricles volume (VEV) were used in this study. These measures were acquired from MRI scans during multiple clinical visits. Whitwell et al. (2001) recommends normalizing these volumetric measures by dividing them by intracranial volume (ICV) to account for head size and inter-image variability, resulting

in proportional brain volume (WBV/ICV), proportional hippocampus volume (HCV/ICV), and proportional ventricles volume (VEV/ICV) measures[25]. While ADNI1 included both 1.5T and 3T scans (with 3T used for 25% of participants), this study exclusively used brain imaging data from 1.5T MRI to minimize potential variation and bias. ADNI1 utilized 1.5T MRI with a 3D T1-weighted sequence and a dual fast spin echo sequence to assess brain pathology and vascular issues[4].

Covariates

Covariates were included in the analyses based on their reported associations in the literature. For instance, several studies have found that carrying at least 1 allele of APOE-ε4 may accelerate the rate of cognitive decline in brain volume[26,27]. Thus, whether the person was a carrier of APOE-ε4 alleles based on the genotyping of blood samples at the screening visit was included as a covariate in all analyses. Demographic characteristics such as sex, baseline age, and years of education were also included.

Statistical analysis

All analyses were conducted with SAS 9.4. Descriptive statistics and bivariate tests between cognitive abilities and potential risk factors at baseline were assessed with ANOVA and chi-square tests at the $\alpha = 0.05$ level. Linear mixed effects models were used to investigate the longitudinal association between structural brain measurements and ADAS-Cog 13 across groups with different cognitive status. The necessity of random intercepts and/or random slopes were assessed by comparing likelihood ratio tests. Different covariance structures (compound symmetry with heterogeneous variances, unstructured, and auto-regressive with and without heterogeneous variances) were assessed as the variance of cognitive scores was not constant over time. Hybrid models with an unstructured covariance matrix and heterogeneous auto-regressive structure for within subject errors were selected as the best fitting models for these data based on likelihood ratio tests, or Akaike information criterion (AIC) and Bayesian information criterion (BIC) statistics.

Three linear mixed effects models with random intercepts and random slopes of time were conducted to test the longitudinal associations between structural brain measurements and ADAS-Cog13 over time across different cognitive groups. Models included time (in months from baseline), baseline age, sex, years of education, and whether the person was an APOE-ε4 carrier. Additionally, a brain volumetric or thickness measurement was included as described in the following paragraphs.

Model 1 included the five cognitive status groups as previously described, along with WBV/ICV, and an interaction between cognitive status and time. This model assumed that both baseline whole brain volume and its longitudinal effect were the same across different groups. To test these assumptions, Model 2 additionally included an interaction between cognitive status and WBV/ICV to assess whether baseline cognitive dysfunctions differed between cognitive groups depending on proportional whole brain volume. Model 3 further extended this by including an interaction term between cognitive status, WBV/ICV, and time, in order to assess whether the effect of WBV/ICV on cognitive changes over time differed between groups. Model 3 was also re-analyzed by replacing WBV/ICV with specific regional brain volume (Model 3a: HCV/ICV, Model 3b: VEV/ICV) or ERC thickness (Model 3c) to assess whether there was an interaction between specific brain regions on cognitive abilities that differed over time.

Lastly, to correct for multiple comparisons, the Benjamini-Hochberg method[28] was used on all primary measures of interest in the regression models. Specifically, the *p*-values for the measures of interest (main effects of cognitive status, brain regions, and their interactions) were ranked in increasing order based on their magnitude, and compared to the adjusted alpha level (defined as: $(i/m)*Q$ where i = rank, m = total number of tests, and $Q = 0.05$) as recommended in the literature. To be conservative, 'm', was defined as the total primary measures of interest across all regression models ($m = 117$).

Results

Descriptive statistics of demographic characteristics and structural brain variables at baseline are provided in Table 1. Subjects with different cognitive status were similar in age but differed in sex and education. Baseline cognitive abilities also differed significantly between cognitive groups: pMCI and AD groups had larger average baseline ADAS-Cog13 scores, reflecting greater impairments. Moreover, structural brain variables revealed significant differences in global and regional brain measurements between groups: on average pMCI and AD groups had smaller WBV and HCV, thinner ERC, and larger VEV at baseline compared with the sCN group ($p < 0.0001$).

Longitudinal association: cognitive change over time

In all three models, ADAS-Cog13 at baseline and over time were higher in pMCI, and AD relative to sCN (Table 2), indicating greater impairments at baseline and faster rates of decline. Participants in the sMCI group also showed greater impairment in their cognitive abilities at baseline compared to sCN, but the rate of their decline was not different. This did not persist after accounting for the interactions between cognitive status, WBV/ICV, and time (Model 3) once multiple comparisons were taken into account. Greater WBV/ICV was associated with slower declines in cognitive abilities over time ($B = -0.01$, $p < 0.0001$, in Modesl 1 and 2). These results persisted after controlling for multiple comparisons.

Although individuals with higher WBV/ICV at baseline exhibited better cognitive performance regardless of their cognitive status ($B = -0.26$, $p < 0.0001$; Model 1), this association was no longer statistically significant once accounting for the interaction between WBV/ICV and cognitive status (Model 2) nor the 3-way interaction between WBV/ICV, cognitive status, and time (Model 3). However, after including the interaction of WBV/ICV and cognitive status in Model 2, participants in the pMCI and AD groups with larger brain volume (WBV/ICV) showed lower cognitive impairment at baseline relative to sCN ($B = -0.28$, $p < 0.05$, and $B = -0.74$, $p < 0.0001$,

Characteristics[a]	sCN ($n=209$)	pCN ($n=18$)	sMCI ($n=206$)	pMCI ($n=171$)	unsMCI ($n=20$)	AD ($n=190$)	p- value
Demographic							
Male, %	51.2	55.6	65.5	61.4	80.0	52.1	0.01
Age, in years	75.7 (5.0)	77.8 (5.4)	74.8 (7.6)	74.7 (7.0)	74.4 (9.3)	75.3 (7.5)	0.35
Education, in years	16.1 (2.9)	15.7 (2.8)	15.5 (3.2)	15.8 (2.9)	15.5 (2.3)	14.7 (3.1)	0.0004
Genetic factor							
APOE-ε4 carrier, %	25.8	33.3	44.2	67.3	30.0	65.8	< 0.0001
Regional brain measurements							
WBV, cm^3	1,006.2 (101.3)	1,014.3 (87.2)	1,007.2 (106.5)	980.2 (112.2)	1,058.5 (113.3)	952.3 (107.7)	< 0.0001
HCV, cm^3	7.3 (0.9)	6.8 (1.0)	6.7 (1.0)	6.0 (1.0)	7.4 (1.1)	5.6 (1.1)	< 0.0001
VEV, cm^3	35.1 (20.1)	37.5 (16.3)	42.6 (24.1)	47.4 (23.1)	37.0 (21.6)	50.3 (25.3)	< 0.0001
ERC, cm	382.9 (64.8)	349.3 (94.3)	350.4 (72.2)	303.1 (70.8)	379.1 (63.0)	273.6 (69.3)	< 0.0001
ICV, cm^3	1,534.4 (157.5)	1,585.7 (157.5)	1,573.3 (159.0)	1,571.6 (177.0)	1,636.1 (195.0)	1,548.4 (181.8)	0.03
ADAS-Cog 13	9.3 (4.2)	12.0 (4.0)	17.1 (6.1)	21.3 (5.3)	11.4 (4.5)	29.0 (7.6)	< 0.0001
Follow-up, in months	38.2 (12.2)	39.7 (5.9)	30.2 (15.0)	35.8 (11.8)	36.3 (10.2)	21.2 (8.6)	< 0.0001

Table 1. Baseline characteristics of participants by cognitive status, ADNI1 ($n=814$). Descriptive statistics and bivariate analyses examining the relationship between cognitive abilities and potential risk factors at baseline were conducted using ANOVA and chi-square tests, with a significance level of $\alpha=0.05$. *sCN* stable-cognitively normal, *pCN* progressed cognitively normal, *sMCI* stable mild cognitive impairment, *pMCI* progressed MCI, *unsMCI* unstable MCI, *AD* stable Alzheimer's disease, *HCV* hippocampus volume, *ERC* entorhinal cortex thickness, *VEV* ventricles volume, *ICV* intracranial volume. [a]Mean (SD) unless noted otherwise.

respectively, Model 2). These groups continued to have faster rates of decline relative to sCN (B = 0.24, $p<0.001$, and B = 0.31, $p<0.0001$, respectively) regardless of their WBV/ICV. After accounting for the multiple testing, the interaction between WBV/ICV and pMCI was no longer significant. Additionally, when including their 3-way interaction, greater WBV/ICV slowed the rate of impairment in cognitive abilities among pMCI and AD participants compared to sCN (Model 3).

Longitudinal association: cognitive change and brain regions

The longitudinal effect of structural brain region measures on ADAS-Cog13 changes is provided in Table 3. At baseline, the sMCI and AD groups showed greater cognitive impairments compared to the sCN group after adjusting for age and other covariates (B = 15.65, $p<0.0001$, and B = 25.01, $p<0.0001$, respectively; Model 3a). However, these impairments could be mitigated by higher HCV/ICV ratios in the sMCI and AD groups (B= − 20.11, $p<0.05$, and B= − 18.88, $p<0.05$, respectively; Model 3a). sMCI and AD participants with thicker ERC also showed fewer impairments (B= − 0.02, $p<0.05$, and B= − 0.02, $p<0.05$, respectively) in cognitive abilities at baseline levels compared to sCN (Model 3c). In contrast, participants with AD who had larger VEV/ICV had significantly worse ADAS-Cog13 scores compared to sCN adjusting for other covariates (B = 0.94, $p<0.05$; Model 3b). The rate of change in cognitive abilities over time based on these brain regions were largely not statistically significant across the different cognitive status groups, relative to sCN ($p>0.05$), except for faster cognitive impairment declines among pMCI subjects with greater VEV/ICV. For the AD group, the interactions between cognitive status and brain region, including HCV/ICV (Model 3a), VEV/ICV (Model 3b), and ERC (Model 3c), were no longer significant after controlling for the multiple comparisons.

Discussion

While the literature has detected potentially differential relationships across brain regions[15,16], this current study specifically further investigated these relationships among subjects with different cognitive statuses at baseline, as well as changing statuses over time. Although cognitive abilities decreased over time in most groups, the rate of decline was fastest for the AD and pMCI groups relative to sCN participants. Brain volume affected the rate of change such that greater proportional brain volume might decrease the impairments or slow down the rate of decline over time in the pMCI and AD groups.

A previous study revealed that both baseline and rate of decline in cognition depended on WBV/ICV in participants with AD, and the impact of brain volume on the rate of decline in cognition among AD individuals was different from other groups[16,29,30]. Indeed, in this current study, AD participants showed decline in their cognitive abilities over time with differing rates depending on their whole brain volume. A previous study by Armstrong et al.[31] in 2020 reported that the faster rate of cognitive decline was correlated with the faster rate of brain atrophy and ventricular enlargement but focused only on cognitively normal subjects. Our study extended the existing literature by separately assessing changes in cognition between those with AD, stable MCI, and those with progressive MCI (cognitively normal at baseline, but progressed to MCI during the follow-up). We were therefore able to detect that whole brain and ventricles volume affected the decrease in cognition among pMCI participants but not among the sMCI patients. However, the effect of ventricles volume was no longer significant after adjusting for multiple comparisons.

Covariates[a]	Model 1; B (95% CI)	Model 2; B (95% CI)	Model 3; B (95% CI)
Cognitive status			
sCN	Reference	Reference	Reference
pCN	1.99 (– 0.68, 4.66)	8.12 (– 33.86, 50.10)	9.59 (– 33.67, 52.85)
sMCI	7.01 (5.91, 8.11)***	20.70 (6.27, 35.13)**	15.77 (0.87, 30.67)*^
pMCI	10.68 (9.46, 11.90)***	28.85 (13.74, 43.96)**	24.30 (8.72, 39.88)**
unsMCI	1.61 (– 0.94, 4.16)	– 9.18 (– 39.23, 20.88)	– 4.66 (– 36.75, 27.43)
AD	18.14 (16.91, 19.37)***	64.47 (48.73, 80.21)***	55.08 (39.36, 70.81)***
Time	0.77 (0.52, 1.02)***	0.77 (0.48, 1.06)***	0.37 (– 0.04, 0.78)
Education	– 0.23 (– 0.37, – 0.09)**	– 0.21 (– 0.35, – 0.07)**	– 0.24 (– 0.38, – 0.10)**
Age (centered)	– 0.01 (– 0.07, 0.05)	– 0.08 (– 0.14, – 0.02)**	– 0.02 (– 0.08, 0.04)
Sex (female vs. male)	– 0.35 (– 1.15, 0.45)	0.04 (– 0.74, 0.82)	– 0.24 (– 1.05, 0.56)
APOE-ε4 carrier	0.75 (– 0.07, 1.57)	1.13 (0.33, 1.93)**	0.81 (– 0.01, 1.62)
WBV/ICV	– 0.26 (– 0.36, – 0.16)***	– 0.02 (– 0.18, 0.14)	– 0.08 (– 0.24, 0.08)
WBV/ICV*Time	– 0.01 (– 0.013, – 0.01)***	– 0.01 (– 0.02, – 0.008)***	– 0.01 (– 0.01, 0.00)
Cognitive status*WBV/ICV			
sCN		Reference	Reference
pCN		– 0.08 (– 0.73, 0.57)	– 0.11 (– 0.78, 0.56)
sMCI		– 0.21 (– 0.43, 0.01)	– 0.13 (– 0.37, 0.10)
pMCI		– 0.28 (– 0.52, – 0.04)*^	– 0.21 (– 0.45, 0.04)
unsMCI		0.17 (– 0.28, 0.62)	0.10 (– 0.39, 0.59)
AD		– 0.74 (– 0.99, – 0.49)***	– 0.59 (– 0.84, – 0.34)***
Cognitive status*Time			
sCN	Reference	Reference	Reference
pCN	0.12 (0.02, 0.22)*	0.11 (– 0.01, 0.23)	– 0.37 (– 1.84, 1.10)
sMCI	0.05 (0.01, 0.09)*^	0.02 (– 0.04, 0.08)	0.38 (– 0.25, 1.01)
pMCI	0.27 (0.23, 0.31)***	0.24 (0.18, 0.30)***	1.20 (0.59, 1.81)**
unsMCI	– 0.04 (– 0.14, 0.06)	– 0.03 (– 0.15, 0.09)	– 1.23 (– 2.52, 0.06)
AD	0.37 (0.31, 0.43)***	0.31 (0.25, 0.37)***	1.67 (0.85, 2.49)***
Cognitive status *WBV/ICV*Time			
sCN			Reference
pCN			0.008 (– 0.02, 0.03)
sMCI			– 0.01 (– 0.02, 0.005)
pMCI			– 0.02 (– 0.03, – 0.001)**
unsMCI			0.02 (– 0.002, 0.04)
AD			– 0.02 (– 0.04, – 0.009)**

Table 2. Results of linear mixed models with random intercept and random effect of time in ADAS-Cog13, exploring the effect of whole brain volume on change in general cognitive abilities over time in participants with different cognitive status. *sCN* stable-cognitively normal, *pCN* progressed cognitively normal, *sMCI* stable mild cognitive impairment, *pMCI* progressed MCI, *unsMCI* unstable MCI, *AD* stable Alzheimer's disease, *WBV* whole brain volume, *ICV* intracranial volume. *p-values from two tailed tests for significance at the $\alpha = 0.05$ level, $p < 0.05$, **$p < 0.01$, ***$p < 0.0001$. ^No longer statistically significant after controlling for multiple comparisons [a]sCN was the reference group in all models.

For some cognitive groups (e.g., pMCI), no associations between the structural brain variables and cognitive abilities were detected, which is in contrast with the literature[32,33]. For instance, we did not find any baseline association between ADAS-Cog13 scores and any brain region explored in this study among pMCI individuals. These findings differ from those of Jack Jr et al. (1999), who reported an association between baseline hippocampus atrophy and the likelihood of progression to AD in individuals with MCI. This inconsistency suggests that the general cognition or its sub-domains may be associated with particular brain regions distinctly over different AD stages. This is further supported by a study in which the authors found that longitudinal changes in hippocampus volume was associated with changes in memory[34]. Indeed, in our study sMCI and AD individuals with greater hippocampus volume and thicker entorhinal cortex displayed less cognitive dysfunction at baseline compared with sCN individuals. Further research is needed to clarify the association between general and sub-domains of cognitive abilities and structural brain variables and their effect on changes in cognition over time across AD stages, particularly at earlier stages. In our study, no significant differences in baseline WBV/ICV or its rate of change were observed in the unsMCI group relative to sCN. Similar patterns emerged when whole brain volume was replaced with hippocampal volume, ventricular volume, or entorhinal cortex thickness. These findings should be interpreted with caution due to the small number of unsMCI participants. Further research

Covariates (1)	Model 3a; B (SE)	Model 3b; B (SE)	Model 3c; B (SE)
Brain region (2)	HCV/ICV	VEV/ICV	ERC
Cognitive status			
sCN	Reference	Reference	Reference
pCN	− 2.83 (− 20.15, 14.48)	− 0.75 (− 7.79, 6.29)	0.65 (− 7.56, 8.86)
sMCI	15.65 (8.32, 22.98)***	6.88 (4.53, 9.23)***	12.85 (8.48, 17.22)***
pMCI	14.65 (7.08, 22.22)**	10.31 (7.62, 13.0)***	13.82 (9.53, 18.11)***
unsMCI	− 2.80 (− 20.73, 15.13)	3.31 (− 2.30, 8.92)	0.26 (− 11.09, 11.61)
AD	25.01 (17.76, 32.26)***	15.61 (12.94, 18.28)***	22.93 (18.54, 27.32)***
Time	0.13 (− 0.09, 0.35)	− 0.04 (− 0.10, 0.02)	0.02 (− 0.14, 0.18)
Education	− 0.22 (− 0.35, − 0.08)**	− 0.22 (− 0.35, − 0.09)**	− 0.19 (− 0.33, − 0.06)**
Age (centered)	− 0.01 (− 0.07, 0.05)	− 0.008 (− 0.07, 0.05)	0.02 (− 0.04, 0.08)
Sex (female vs. male)	− 0.32 (− 1.17, 0.53)	− 0.13 (− 0.96, 0.70)	− 1.08 (− 1.92, − 0.24)*
APOE-ε4 carrier	0.74 (− 0.14, 1.61)	0.82 (− 0.02, 1.65)	0.86 (0.004, 1.71)*
Cognitive status*Time			
sCN	Reference	Reference	Reference
pCN	0.47 (− 0.13, 1.07)	0.10 (− 0.15, 0.35)	0.39 (0.03, 0.75)*^
sMCI	0.14 (− 0.15, 0.44)	0.02 (− 0.08, 0.11)	− 0.06 (− 0.27, 0.15)
pMCI	0.16 (− 0.13, 0.46)	0.17 (0.06, 0.27)**	0.31 (0.12, 0.50)**
unsMCI	− 0.21 (− 1.05, 0.63)	0.04 (− 0.16, 0.24)	0.09 (− 0.47, 0.65)
AD	0.07 (− 0.26, 0.41)	0.32 (0.19, 0.45)***	0.28 (0.05, 0.51)*^
Brain region (2)	− 8.12 (− 19.68, 3.44)	0.69 (0.02, 1.36)*^	− 0.003 (− 0.01, 0.006)
Brain region*Time (2)	− 0.30 (− 0.77, 0.17)	0.02 (0.006, 0.04)*	− 0.0001 (− 0.0004, 0.0003)
Cognitive status*Brain region (2)			
sCN	Reference	Reference	Reference
pCN	11.90 (− 26.99, 50.79)	1.36 (− 1.44, 4.16)	0.01 (− 0.01, 0.03)
sMCI	− 20.11 (− 36.16, − 4.06)*	0.11 (− 0.73, 0.95)	− 0.02 (− 0.04, − 0.004)**
pMCI	− 10.15 (− 27.87, 7.57)	0.27 (− 0.65, 1.19)	− 0.01 (− 0.02, 0.004)
unsMCI	9.19 (− 29.68, 48.06)	− 0.69 (− 2.90, 1.52)	0.003 (− 0.03, 0.03)
AD	− 18.88 (− 35.93, − 1.83)*^	0.94 (0.06, 1.82)*^	− 0.02 (− 0.03, − 0.002)*^
Cognitive status*Brain region*Time (2)			
sCN	Reference	Reference	Reference
pCN	− 0.82 (− 2.25, 0.61)	0.01 (− 0.07, 0.09)	− 0.001 (− 0.003, 0.001)
sMCI	− 0.27 (− 0.94, 0.40)	0.01 (− 0.03, 0.05)	0.0004 (− 0.0002, 0.001)
pMCI	0.35 (− 0.39, 1.09)	0.03 (0.004, 0.07)*^	0.00002 (− 0.001, 0.001)
unsMCI	0.43 (− 1.43, 2.29)	− 0.03 (− 0.11, 0.05)	− 0.0003 (− 0.002, 0.002)
AD	0.87 (− 0.01, 1.75)	0.02 (− 0.02, 0.06)	0.001 (− 0.0002, 0.002)

Table 3. Results of linear mixed models with random intercept and random effect of time in ADAS-Cog13, exploring the effect of structural brain measures on change in general cognitive abilities over time in participants with different cognitive status. *sCN* stable-cognitively normal, *pCN* progressed cognitively normal, *sMCI* stable mild cognitive impairment, *pMCI* progressed MCI, *unsMCI* unstable MCI, *AD* stable Alzheimer's disease, *ICV* intracranial volume, *HCV/ICV* proportional hippocampus volume to ICV, *ERC* entorhinal cortex thickness, *VEV/ICV* proportional ventricles volume to ICV. *p-values from two tailed tests for significance at the $\alpha = 0.05$ level, $p < 0.05$, ** $p < 0.01$, *** $p < 0.0001$. ^No longer statistically significant after controlling for multiple comparisons. (1) sCN was the reference group in all models. (2) Brain region is HCV/ICV in Model 3a; VEV/ICV in Model 3b; ERC in Model 3c.

is needed to explore the relationship between cognitive abilities and structural brain variables among subjects in this subgroup.

These findings support the brain reserve hypothesis, suggesting that greater brain volume may slow cognitive decline in AD and pMCI[35]. However, the inconsistent associations between brain measures and cognition, especially in pMCI, highlight the role of cognitive reserve. This theory suggests that individuals with similar brain pathology may have different cognitive outcomes due to compensatory mechanisms, explaining why certain brain regions were associated with cognitive function at baseline but not with the rate of decline[36].

This study has some strengths. We used ADAS-Cog13 which better addresses changes in general cognitive abilities over time compared to other tools commonly used in the literature such as MMSE[2]. Similarly, due to the ADNI protocol of follow-up every six months, the assessment of cognitive *status* was relatively precise, and the analysis could also incorporate whether it was changing over time. This enabled us to identify distinctive groups

while accounting for their baseline cognitive status as well as changes in cognition over time. Secondly, we directly investigated the relationship between global and regional brain measurements with cognitive abilities at baseline, over time, as well as their rate of change. Our analytic approach incorporated brain imaging, physician's assessment, and validated questionnaire responses at baseline alongside their follow-up measures to classify participants into mutually exclusive stable or changing cognitive status groups.

This study also has some limitations. It was only focused on the change in general cognitive abilities and brain atrophy. In contrast, some studies showed the association between domain-specific cognitions and regional brain atrophy. For instance, Armstrong et al. (2020)[31] found that different regional brain changes were associated with different cognitive domains. The authors warn against using general cognition as a measure to study changes in different brain regions and their association with changes in cognitive abilities. While ADNI1 mainly implemented 1.5T MRI scans, image scans in subsequent ADNI phases were 3T, which is preferable due to the larger signal to noise ratio[4,37]. However, results should be comparable, with a low risk of misclassification[38]. Results from this study are in their original units, rather than after standardization. As ADAS-Cog13 has a well-defined clinical interpretation, we made this analytic choice to preserve the clinical relevancy of the findings.

This study had a small number of people who were cognitively normal at baseline and progressed to MCI or AD during follow-up, as well as a small number of people who were MCI at baseline and subsequently progressed to AD and reverted back to MCI during follow-up; results for these two subgroups should be interpreted with caution. Nevertheless, the sample size was large ($N = 814$, with ~200 participants in each of the stable cognition categories, and ~200 participants who progressed overtime). As there were a large number of comparisons made in this study, we applied corrections for multiple comparisons to mitigate the potential for inflated Type I errors. Notably, most results remained consistent and significant after accounting for the multiple comparisons. The largest discrepancies were those in the AD group. However, the average follow-up time in the AD group was shorter compared to the other groups. This was by design from ADNI, as participants in the AD group were only followed for two years, which is approximately one less year of follow-up compared to the other cognitive groups. Caution is warranted when considering the results before, and after the correction for multiple comparisons for the AD group. While multiple testing raises the risk of Type I errors, over correction may raise the risk of Type II errors. Thus, the study results reported here may be underestimated. Lastly, ADNI is not a representative cohort, and results may not be generalizable to the general population.

Conclusions

This study is among the largest longitudinal studies to investigate the role of brain volumetric measures and their changes over time on cognitive abilities. The rate of decline in cognitive abilities was accelerated with larger decreases in whole brain volume and ventricle enlargement. In contrast, brain regions and their changes over time were not predictive of impairment for the sMCI cognition group. By being able to accurately identify those at highest risk for progression from MCI to AD, results will have clinical implications on the development of interventions to prevent or slow down the progression of MCI. Results highlight the importance of different brain regions on cognition particularly among the MCI group and future research should further investigate these differential effects, especially among those who revert or have fluctuating cognitive statuses over time.

Data availability

Data used in the preparation of this article were obtained from the Alzheimer's Disease Neuroimaging Initiative (ADNI) database (adni.loni.usc.edu). Restrictions apply to the availability of these data, which were used under license for the current study, and so are not publicly available. Data are however available from ADNI (adni.loni.usc.edu), or from the corresponding author (lisa.kakinami@concordia.ca) upon reasonable request and with permission of ADNI.

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Author contributions

Author contributions included conception and study design (FH and LK), statistical analysis (FH), interpretation of results (FH, LK), drafting the manuscript work (FH) or revising it critically for important intellectual content (LK) and approval of the final version to be published and agreement to be accountable for the integrity and accuracy of all aspects of the work (FH and LK).

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Declarations

Competing interests
The authors declare no competing interests.

Ethics approval and consent to participate
Data were obtained through the ADNI data repository (adni.loni.usc.edu). Written informed consent was obtained from all participants and prior institutional review board approval was obtained from all participating institutions. As multiple universities and institutions are heavily involved with ADNI data collection and study management, their full details are omitted from this manuscript. They can be found in the ADNI protocol and ADNI consent forms on the publicly accessible ADNI website: https://adni.loni.usc.edu/methods/documents/. This secondary data analysis was approved by Concordia University's Human Research Ethics Committee (#30019564). De-identified data are publicly available upon approval from the ADNI Data Sharing and Publications Committee.

Consent for publication
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